

GÖKHAN TUTKU ÇAY

Senior Game Engineer & Technical Lead (Gameplay & Multiplayer)

Unity • Real-Time Systems • AAA-Scale Architecture • System Optimization

📍 Turkey – Remote Friendly | ✈️ No Travel Restrictions |

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ABOUT ME & VISION

The unique rhythm of software, mathematics, philosophy, and music is more than just a passion; it is the fundamental cornerstone that shapes my professional vision. I embrace a working principle that merges analytical thinking with the discipline of mathematics, system design with the inquisitive depth of philosophy, and the harmony of code with the rhythm of music.

By centering my professional focus entirely on problem-solving and optimization, I view analyzing every technical challenge deeply and conducting an aesthetic solution production process as an **"intellectual pleasure."** This multifaceted perspective is the primary source of my motivation to develop sustainable, stable, and high-performance systems.

TECHNICAL SKILLS

Gameplay & AI Engineering

- **Core Systems:** Advanced Player Controllers (Physics-based), Combat & Ability Frameworks, Inventory & Interaction Systems.
- **AI & Logic:** NPC Behavior Systems, Finite State Machines (FSM), Behavior Trees, Unity NavMesh, Logic-driven Systems.
- **Architecture:** ScriptableObject-driven design, **Entity Component System (ECS)**, Data-Oriented Design (DOD).

Multiplayer & Real-Time Networking

- **Technology:** Unity Photon (PUN/Fusion), Real-time Network Synchronization.
- **Network Logic:** Matchmaking Algorithms, Latency Handling, Client-side Prediction, Competitive (1v1, 3v3) Architecture.

Performance & System Operations (ITIL & SDLC)

- **Optimization:** Unity Profiler, Memory Management (GC reduction), Mobile Performance Tuning.
- **Operations: ITIL Principles**, SLA Management, Root Cause Analysis (RCA).
- **Pipelines:** Custom Unity Editor Tools, Level Generators, Jenkins (CI/CD basics), Steamworks Integration.

Programming & Foundations

- **Languages:** C# (Expert), C++, C, JavaScript (Node.js), SQL.
- **Engineering:** SOLID, Design Patterns (Factory, Observer, MVC, Command), Microservices, Multithreading.
- **Methodology:** Agile/Scrum Expert (5+ years), SDLC Management, Technical Debt Reduction.

PROFESSIONAL EXPERIENCE

Independent Software Developer & Technical Lead | Remote

2024 – Present

- **Product Life Cycle (Fuu - Steam):** Managed the end-to-end development, QA, optimization, and successful deployment of **"Fuu"** on the Steam platform. Oversaw the entire **Release Management** process and post-launch technical sustainability.
- **Rapid Prototyping (50+ Projects):** Developed 50+ technical prototypes by analyzing market trends; accelerated the idea-to-product conversion time by 30% using modular, reusable code architectures.
- **High-Performance Architecture:** Designed decoupled, event-driven gameplay systems that handle complex physics and environment interactions with 60 FPS stability on target devices.

Game Facto — Senior Game Developer | Istanbul

2022 – 2024

- **Performance Engineering:** Optimized complex system mechanics to maintain system stability and **SLA Compliance**.
- **Root Cause Analysis (RCA):** Actively analyzed critical system errors, implementing permanent solutions that reduced error recurrence rates by 40%.
- **Technical Mentorship:** Provided mentorship to the team on code quality and best practices, ensuring the reduction of technical debt and establishing sustainable code architecture.

Dark Zone — Team Leader & Project Manager | Istanbul

2021 – 2022

- **Team Leadership:** Coordinated an 8-person developer team, ensuring projects were delivered in accordance with technical standards and tight schedules.
- **Incident & Change Management:** Managed cross-team coordination during critical system outages (incidents) and performed risk analysis for changes in live systems.
- **Stakeholder Management:** Acted as a bridge between software and business units (Cross-functional collaboration) to ensure technical requirements were 100% aligned with business goals.

EDUCATION & LANGUAGES

Gaziantep University | Turkish Language & Literature (Undergraduate - Focused on Logic & Systems)

- *Note: Transitioned to full-time Software Engineering to leverage analytical and linguistic logic in complex system architectures.*

Languages: Turkish (Native), English (Professional Working Proficiency).

INTERESTS & COMPETENCIES

- **Social:** Transforming complex technical issues into strategic data and reports for business stakeholders.
- **Personal:** Philosophy of Systems, Music Theory (Rhythm & Harmony), Mathematical Patterns in Nature.